

Q1 - 24 June - Shift 1

Nucleus A is having mass number 220 and its binding energy per nucleon is 5.6 MeV. It splits in two fragments 'B' and 'C' of mass numbers 105 and 115. The binding energy of nucleons in 'B' and 'C' is 6.4 MeV per nucleon. The energy Q released per fission will be :

- (A) 0.8 MeV (B) 275 MeV
(C) 220 MeV (D) 176 MeV

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Q2 - 24 June - Shift 2

A sample contains 10^{-2} kg each of two substances A and B with half lives 4 s and 8 s respectively.

The ratio of then atomic weights is 1 : 2. The ratio

of the amounts of A and B after 16 s is $\frac{x}{100}$. the

value of x is _____.

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Q3 - 26 June - Shift 1

How many alpha and beta particles are emitted

when Uranium ${}_{92}\text{U}^{238}$ decays to lead ${}_{82}\text{Pb}^{206}$?

- (A) 3 alpha particles and 5 beta particles
(B) 6 alpha particles and 4 beta particles
(C) 4 alpha particles and 5 beta particles
(D) 8 alpha particles and 6 beta particles

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Q4 - 26 June - Shift 2

A radioactive nucleus can decay by two different processes. Half-life for the first process is 3.0 hours while it is 4.5 hours for the second process.

The effective half-life of the nucleus will be :

- (A) 3.75 hours (B) 0.56 hours
(C) 0.26 hours (D) 1.80 hours

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Q5 - 26 June - Shift 1

The Q-value of a nuclear reaction and kinetic energy of the projectile particle, K_p are related as :

- (A) $Q = K_p$ (B) $(K_p + Q) < 0$
(C) $Q < K_p$ (D) $(K_p + Q) > 0$

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Q6 - 28 June - Shift 2

Questions

MathonGo

Following statements related to radioactivity are given below:

(A) Radioactivity is a random and spontaneous process and is dependent on physical and chemical conditions.

(B) The number of un-decayed nuclei in the radioactive sample decays exponentially with time.

(C) Slope of the graph of $\log_e(\text{no. of undecayed nuclei})$ Vs. time represents the reciprocal of mean life time (τ).

(D) Product of decay constant (λ) and half-life time ($T_{1/2}$) is not constant.

Choose the most appropriate answer from the options given below:

(A) (A) and (B) only

(B) (B) and (D) only

(C) (B) and (C) only

(D) (C) and (D) only

Q7 - 29 June - Shift 1

The activity of a radioactive material is 2.56×10^3 Ci. If the half life of the material is 5 days, after how many days the activity will become 2×10^5 Ci?

(A) 30 days

(B) 35 days

(C) 40 days

(D) 25 days

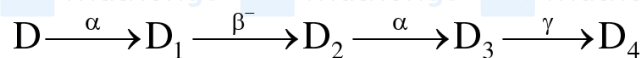
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Q8 - 29 June - Shift 2

In the following nuclear reaction,



Mass number of D is 182 and atomic number is 74.

Mass number and atomic number of D_4 respectively will be ____.

- (A) 174 and 71 (B) 174 and 69
(C) 172 and 69 (D) 172 and 71

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Q9 - 29 June - Shift 2

The half life of a radioactive substance is 5 years.

After x years a given sample of the radioactive substance gets reduced to 6.25% of its initial value of x is ____.

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Answer Key

Q1 (D)

Q2 (25)

Q3 (D)

Q4 (D)

Q5 (D)

Q6 (C)

Q7 (B)

Q8 (A)

Q9 (20)

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Q1 (D)

$$\begin{aligned}
 Q &= (B.E)_P - (B.E)_R \\
 &= (105 + 115)(6.4) - (220)(5.6) \\
 &= 176 \text{ MeV}
 \end{aligned}$$

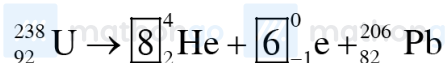
Q2 (25)

$$N_t = N_0 (0.5)^{\frac{t}{t_{1/2}}}$$

$$= \frac{m}{M} \times N_A (0.5)^{\frac{t}{t_{1/2}}}$$

$$\frac{N_1}{N_2} = \frac{M_2}{M_1} (0.5)^{t \left[\frac{1}{T_A} - \frac{1}{T_B} \right]}$$

$$= 2(0.5)^{16 \times \frac{1}{8}} = \frac{2}{4} = \frac{1}{2} = \frac{x}{100}$$

Q3 (D)

8 α particles and 6 β particles are emitted.

Q4 (D)

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$$\lambda_{\text{eq}} = \lambda_1 + \lambda_2$$

$$\frac{\ln 2}{(t_{1/2})_{\text{eq}}} = \frac{\ln 2}{(t_{1/2})_1} + \frac{\ln 2}{(t_{1/2})_2}$$

$$(t_{1/2})_{\text{eq}} = \frac{(t_{1/2})_1 \times (t_{1/2})_2}{(t_{1/2})_1 + (t_{1/2})_2}$$

$$= \frac{3 \times 4.5}{3 + 4.5} = \frac{3 \times 4.5}{7.5} = \frac{3 \times 3}{5} = 1.8 \text{ hr}$$

Q5 (D)



$$Q = k_{\gamma} + k_b - k_p$$

$$Q + k_p = k_{\gamma} + k_b$$

$$Q + k_p > 0$$

Q6 (C)

(C)

Q7 (B)

$$\frac{A}{A_0} = \frac{N}{N_0}$$

$$\frac{2 \times 10^{-5}}{2.56 \times 10^{-3}} = \frac{N}{N_0}$$

$$\frac{N}{N_0} = \frac{1}{128} \Rightarrow N = \frac{N_0}{128}$$

After 7 half life activity comes down to given value $T = 7 \times 5$

= 35 days

Q8 (A)

Say for D_4 Atomic No = Z

Mass Number = A

$$A = 182 - 4 - 4 = 174$$

$$Z = 74 - 2 + 1 - 2 = 71$$

Q9 (20)

$T_{1/2} = 5$ year

$$N = N_0 \left(\frac{1}{2} \right)^{\text{No of half lives}}$$

$$\frac{N}{N_0} = \frac{1}{16} = \left(\frac{1}{2} \right)^4$$

Time = 4 half lives = 20 years